

IET, DAVV, Indore
B.E. I year (IT-A+B)
Class Test – I (Oct. -2023)
1AMRC1: Applied Mathematics-I

Time: 70 min.

Maximum Marks: 20

Note: Attempt only four questions.

- Q:1 If $y = (\sin^{-1}x)^2$, Prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. 5M
- Q:2 Expand $\tan^{-1}x$ in positive power of $(x-1)$ upto first four non-zero terms. 5M
- Q:3 Find that the radius of curvature at a point of the curve $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$. 5M
- Q:4 Evaluate: i) $\int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} dx$ ii) $\int_0^1 \frac{dx}{\sqrt{(1-x^n)}}$ 5M
- Q:5 Trace the curve $y^2(a-x) = x^2(a+x)$. Write Steps properly. 5M